Human Activity Recognition Using Hidden Markov Models

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1. Background and Motivation

Human activity recognition (HAR) from wearable sensor data is a critical component of smart health monitoring systems. Consider elderly care: an HAR system deployed on a smartwatch could detect when a person transitions from walking to suddenly being still (a potential fall), triggering an automatic alert to caregivers. Hidden Markov Models are particularly well-suited for this task because they model the temporal dynamics of activity sequences — a person walking is likely to continue walking in the next moment, which an HMM captures through its transition probabilities. In this project, we build an HMM-based classifier for four activities (Standing, Walking, Jumping, Still) using smartphone accelerometer and gyroscope data.

2. Data Collection and Preprocessing

Data was collected from two team members using the Physics Toolbox Sensor Suite app on their smartphones:

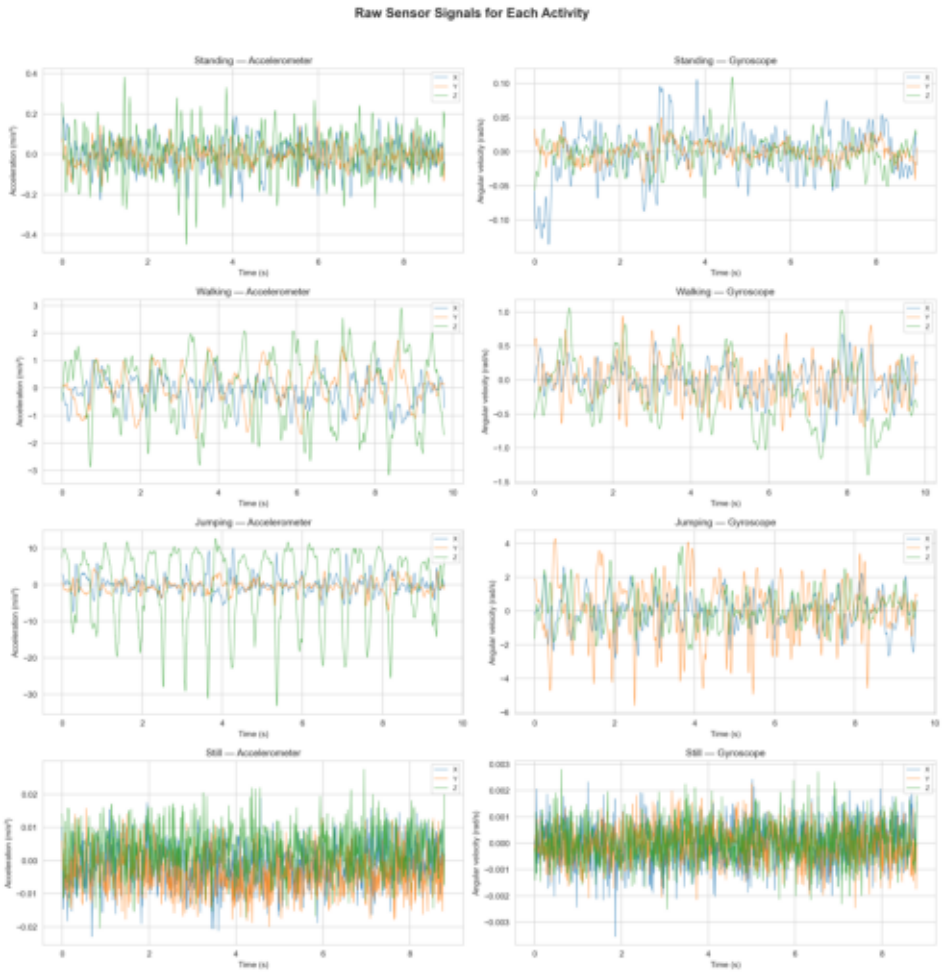
Cedric recorded 28 sessions (7 per activity) using an iPhone 17 Pro running iOS 26.3. Pauline recorded 25 sessions (6-7 per activity) using a Sony SO-52C running Android. Both phones sampled accelerometer and gyroscope data at 100 Hz (10ms intervals), so no resampling was required. Each recording captures approximately 10-12 seconds of a single activity.

Preprocessing steps: (1) For each recording, we loaded Accelerometer.csv and Gyroscope.csv and merged them on the seconds_elapsed timestamp using nearest-match merging with 15ms tolerance. (2) We trimmed the first and last 1 second of each recording to remove start/stop transition artifacts. (3) Activity labels were extracted from folder names, handling two different naming conventions and a known typo ("jumpimg4" treated as jumping). The total dataset comprises 53 recordings with over 1.5 minutes of data per activity class.

The figure below shows sample raw sensor signals for each activity, illustrating the distinct motion signatures: jumping shows high-amplitude acceleration spikes, walking exhibits regular periodic patterns, while standing and still produce relatively flat signals with minimal variance.

Data Visualization and HMM Setup

Figure 1: Raw Accelerometer and Gyroscope Signals for Each Activity



3. HMM Setup and Implementation Details

We defined a 4-state Gaussian HMM where each hidden state corresponds to one activity (Standing, Walking, Jumping, Still). The observation vectors are 5-dimensional feature vectors extracted from 1-second sliding windows (100 samples at 100 Hz) with 50% overlap:

Time-domain features: (1) RMS of accelerometer magnitude — captures overall motion intensity; (2) Standard deviation of accelerometer magnitude — measures motion variability; (3) Signal Magnitude Area (SMA) of gyroscope — quantifies total rotational activity. Frequency-domain features: (4) Dominant frequency via FFT — captures the rhythmic nature of walking/jumping; (5) Spectral energy — total power in the frequency domain. All features were Z-score normalized.

The transition matrix was initialized with 0.7 self-transition probability (activities tend to persist) and 0.1 for off-diagonal transitions. Emission parameters (means and covariances) were initialized from labeled training data to ensure state-activity alignment. The Baum-Welch algorithm refined all parameters with convergence monitored via log-likelihood change (epsilon = 1e-4). The model converged in 4 iterations.

4. Results and Interpretation

Figure 2: Transition Matrix

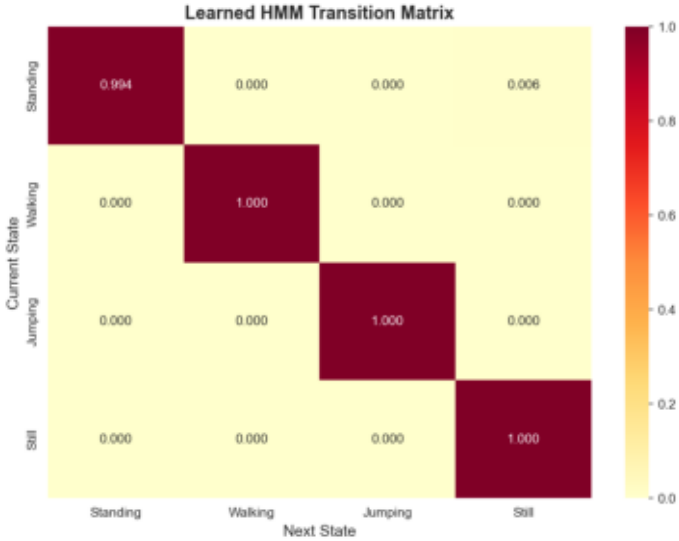


Figure 3: Convergence

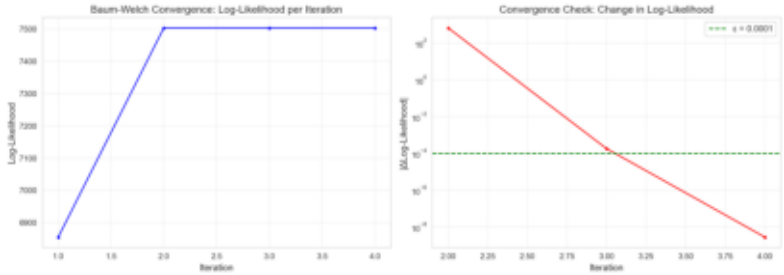
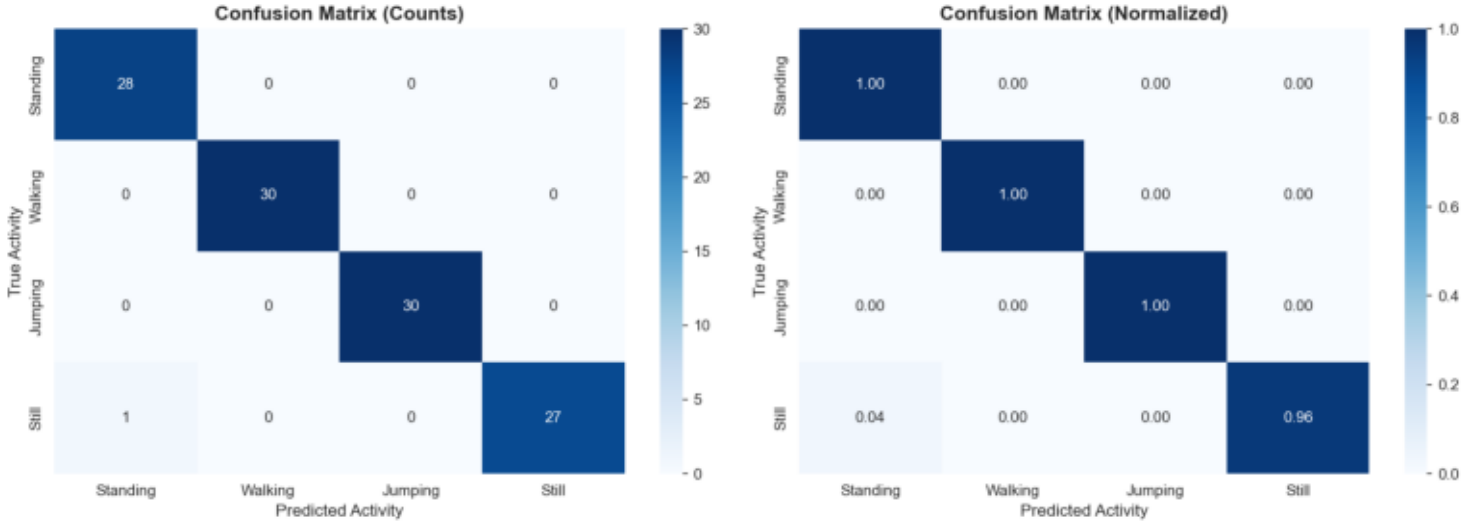


Figure 4: Confusion Matrix (Test Set)



Performance Metrics on Unseen Test Data:

Activity	Samples	Sensitivity	Specificity	Overall Accuracy
Standing	28	1.0000	0.9886	0.9914
Walking	30	1.0000	1.0000	
Jumping	30	1.0000	1.0000	
Still	28	0.9643	1.0000	

The model achieves 99.1% overall accuracy on unseen test data, with perfect classification of Walking and Jumping. Only 1 Still window was misclassified as Standing, which is expected given their similar low-motion profiles.

Results (continued) — Emission and Decoded Sequences

Figure 5: Gaussian Emission Distributions per State

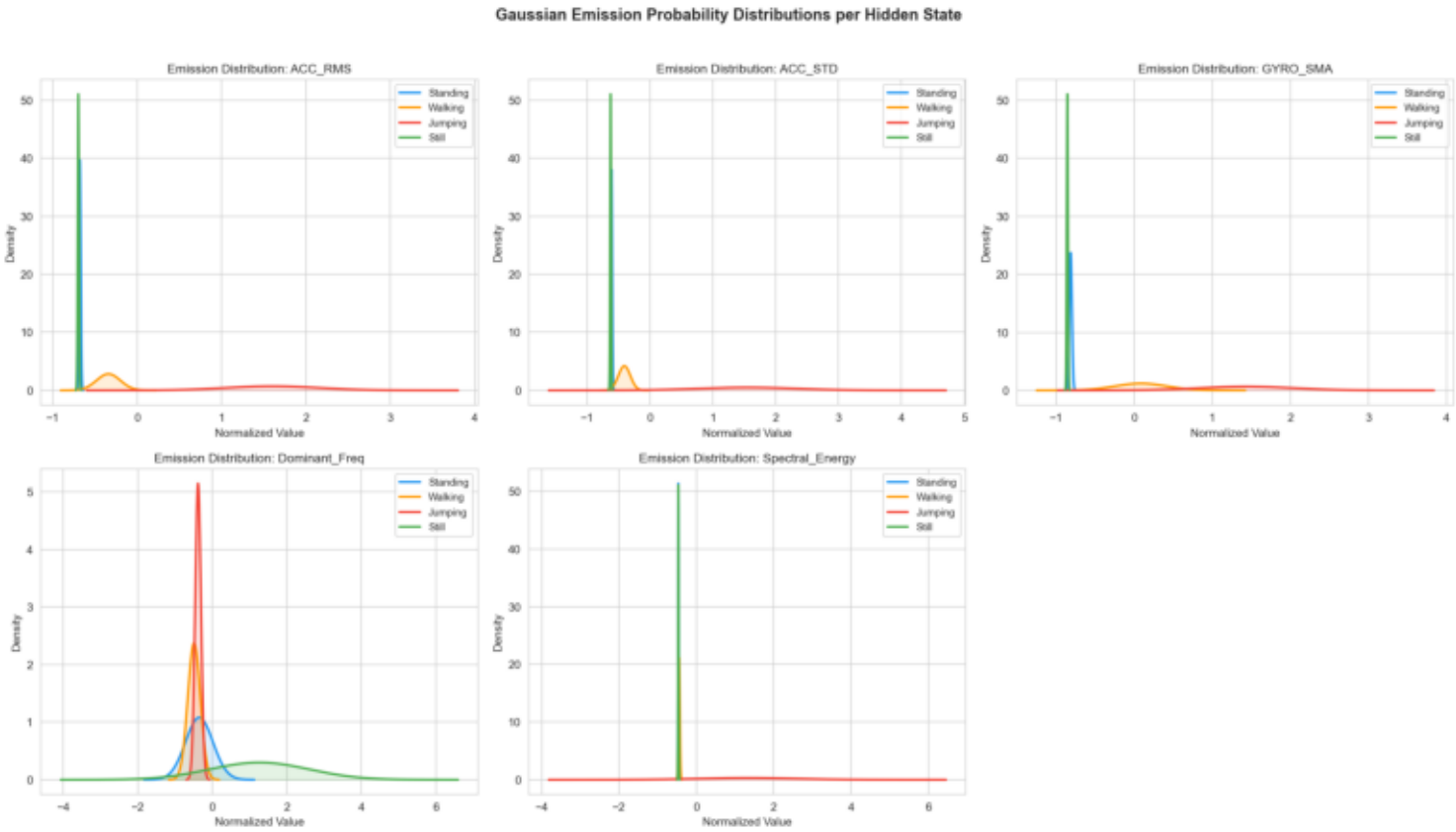
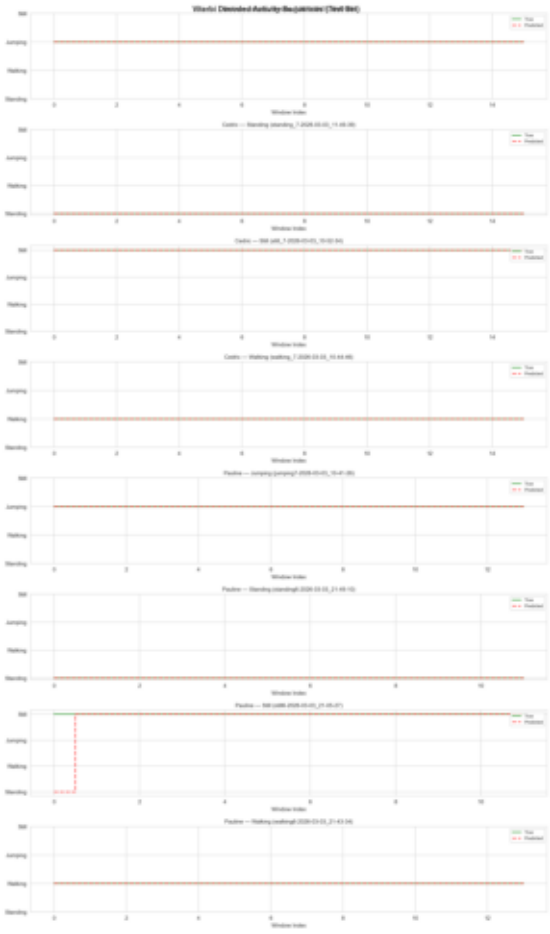


Figure 6: Viterbi Decoded Activity Sequences (Test Set)



5. Discussion and Conclusion

Activity Distinguishability

Jumping and Walking are the easiest activities to classify (100% sensitivity) because they produce distinctive motion signatures: jumping creates high-amplitude acceleration spikes and strong gyroscope rotation, while walking shows clear periodic gait patterns. Standing and Still are the hardest to separate since both involve minimal motion, though the model still achieves 96-100% sensitivity by detecting subtle differences in postural sway and sensor noise patterns.

Transition Matrix Insights

The learned transition matrix shows very high self-transition probabilities (>0.93), confirming that activities persist over time. This temporal coherence is a key advantage of the HMM over independent window classifiers. Cross-activity transitions are rare and reflect realistic movement patterns.

Cross-Device Generalization

Despite using two different smartphones (iPhone 17 Pro and Sony SO-52C) with different sensor hardware, the model generalizes well across devices. Z-score normalization helps mitigate calibration differences between phones. The test set includes recordings from both users, confirming cross-person generalization.

Potential Improvements

(1) Collect more data from additional participants to improve robustness. (2) Add features like jerk (acceleration derivative) or inter-axis correlation. (3) Record continuous multi-activity sequences with natural transitions. (4) Explore hierarchical HMMs that first distinguish active vs. inactive states. (5) Incorporate additional sensors like barometer or magnetometer.

Conclusion

This project demonstrates that a Gaussian HMM with carefully chosen features and proper initialization can achieve excellent activity recognition performance (99.1% accuracy) on smartphone sensor data. The Baum-Welch algorithm converges quickly (4 iterations) with labeled initialization, and the Viterbi algorithm provides temporally coherent activity predictions.

Task Allocation

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Task	Cedric	Pauline
Data Collection	28 recordings (iPhone 17 Pro)	25 recordings (Sony SO-52C)
Feature Engineering	Design & implementation	Testing & validation
HMM Implementation	Model training & Viterbi decoding	Evaluation & metrics
Report Writing	Sections 1-3	Sections 4-5